Accuracy of Pulse Oximeters Intended for Hypoxemic Pediatric Patients

Bronwyn U. Harris, MD[1]; Danton S. Char, MD[2]; Jeffrey A. Feinstein, MD[1]; Archana Verma, MBBS[2]; Stephen C. Shiboski, PhD[3]; Chandra Ramamoorthy, MD[2]

Objectives: Prior studies have shown inaccuracies in pulse oximetry readings at saturations less than 85%; however, no large studies have evaluated new sensors marketed for these low saturations. This study's purpose was to evaluate two sensors with claims of improved accuracy in children with saturations less than 85%.

Design: Prospective observational study.

Setting: Single institution; cardiac catheterization laboratory, and operating room.

Patients: Fifty patients weighing 3–20 kg with baseline saturations less than 90% undergoing surgical or catheterization procedure.

Measurements and Main Results: Data collected included demographics, diagnosis, continuous saturations from three different pulse oximeters (Masimo LNCS [Masimo, Irvine, CA], Masimo Blue [Masimo], and Nellcor Max-I [Medtronic, Dublin, Ireland]) and up to four blood samples for co-oximetry as the gold-standard arterial oxygen saturation. Analysis included scatter plots, smoothed regression estimates of mean continuous saturation levels plotted against corresponding arterial oxygen saturation values, and Bland-Altman plots. Bland-Altman analysis indicated increasing levels of bias and variability for decreasing arterial oxygen saturation levels for all three sensors, with a statistically significant increase in mean difference observed for decreasing arterial oxygen saturation level. The Masimo Blue sensor had the lowest mean difference, SD and Bland-Altman limits in patients with saturations less than or equal to 85%. At saturation range of less than or equal to 85% and greater than 75%, 14% of the samples obtained from Masimo Blue, 24% of the readings from the Nellcor, and 31% from the Masimo Standard sensors were greater than or equal to 5% points difference. All three sensors had a further increase in these differences for arterial oxygen saturation values less than 75%.

Conclusions: The Masimo Blue sensor has improved accuracy at saturations 75–85% versus the Nellcor and Masimo Standard sensors. The accuracy of peripheral capillary oxygen saturation of the Masimo Blue sensor was within 5% points of the arterial oxygen saturation the majority of the time. Currently, at saturations less than or equal to 85%, pulse oximetry alone should not be relied on in making clinical decisions. (*Pediatr Crit Care Med* 2016; 17:315–320)

Key Words: congenital heart disease/defects; hypoxia; monitoring; pediatric; pulse oximetry

[1]Department of Pediatrics (Cardiology), Stanford University School of Medicine, Palo Alto, CA.

[2]Department of Anesthesiology and Critical Care (Pediatric Anesthesiology), Stanford University School of Medicine, Palo Alto, CA.

[3]Department of Epidemiology and Biostatistics, University of California, San Francisco, CA.

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For information regarding this article, E-mail: buharris@stanford.edu

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Pulse oximetry has been in clinical use since the 1970s and has become an integral part of patient care (1). It provides a continuous noninvasive measurement of arterial oxygen saturation (Sa_{O_2}) and is used as both a diagnostic and prognostic monitor in the management of pediatric and adult patients.

Pulse oximetry directly measures transmittance of two wavelengths of light, differentially absorbed by deoxygenated and oxygenated hemoglobin. This transmittance is then converted into a saturation percentage, using a calibration graph or algorithm based on co-oximetry, the gold-standard measurement of Sa_{O_2}. The calibration, however, is most often performed on healthy adult volunteers, a population with very different characteristics than, for example, cyanotic pediatric patients (2–4).

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It should not be surprising then that pulse oximeters have been shown in studies to be poorly calibrated for low saturations (5).

In children with cyanotic forms of congenital heart disease, the arterial saturations may consistently lie between 75% and 85%. In these children, clinical management decisions regarding admission, oxygen administration, or timing of cardiac catheterization or surgery are often made based on pulse oximetry readings. However, a recent large multi-institutional study in critically ill children demonstrated significant inaccuracies in the 80–85% saturation range (6). This study was conducted using standard of care sensors not specifically made for hypoxemic patients. There are now pulse oximeters specifically marketed for patients with low saturations that are purported to be more accurate in these ranges. The data, however, are quite limited, with the largest study to date including only 12 patients (7–9). Furthermore, these sensors can cost twice as much as the standard sensors, raising issues of cost effectiveness.

The aim of the present study was to determine, in children with cyanotic congenital heart disease, which of the three sensors, Masimo SET with LNCS sensor (Masimo Standard; Masimo, Irvine, CA), Masimo SET with blue sensor (Masimo Blue; Masimo), or Nellcor N-600 with Max-I sensor (Nellcor; Medtronic, Dublin, Ireland), is more accurate at lower saturations when compared with arterial co-oximetry. The three sensors in the study have different marketed accuracy claims, but all claim to be within 3% points when saturations are above 80% and no more than 4% points difference at lower saturations as compared with co-oximetry.

We evaluated whether each of these sensor type's inaccuracies at low saturations were within a clinically acceptable range for patient care of patients with saturations persistently 75–85%, which we define as less than or equal to 5% points difference in pulse oximetry reading (peripheral capillary oxygen saturation [Sp_{O_2}]) and the gold-standard co-oximetry (Sa_{O_2}), a determination based on prior published use of pulse oximetry in children and the clinical use of pulse oximetry at our institution (10, 11).

METHODS

Patients and Data Collection

This investigator-initiated prospective observational study was performed at Lucile Packard Children's Hospital at Stanford in the cardiac catheterization laboratories and the cardiac operating rooms (ORs) between November 2012 and September 2014. Following institutional review board and written parental approval, infants and children weighing 3–20 kg with saturations at or below 90% (measured by pulse oximetry at time of assessment in the clinic) were enrolled in the study. Exclusion criteria were existence of known hemoglobinopathies, emergency procedures, or absence of arterial access. The trial was registered on ClinicalTrials.gov, number NCT02237014.

Patients were premedicated when appropriate and monitors were placed, including electrocardiogram, blood pressure, and the pulse oximeter (Masimo Standard). The Masimo Standard pulse oximeter(s) was placed at the discretion of the physician and as result the location varied. Anesthesia was then administered at the discretion of the perioperative physician caring for the patient. Frequently, two Masimo Standard sensors were placed as standard of care. Therefore, for an individual patient, up to two values from Masimo Standard sensors could be obtained with each co-oximetry value. An arterial line was placed as part of clinical care for perioperative monitoring and/or as part of the procedure in the catheterization laboratory or operating room. Following this standard of care monitoring, two additional test sensors (Masimo Blue and Nellcor) were placed on a thumb or great toe or if those locations were not available then placed on a finger. To avoid issues of hypoperfusion-related artifact, study sensors were exclusively placed on an extremity without an arterial line. Pulse oximeter values for the standard and study sensors were recorded continuously and stored securely for off-line data analysis. For the purposes of the study, each subject had a maximum of four arterial samples drawn to minimize the blood withdrawn as part of the study. Blood samples were attempted to be drawn when oxygen saturation levels were stable. The blood sample was immediately processed using a co-oximeter (AVOXimeter 1000E, Edison, NJ). Although attempts were made to collect four study samples from each patient, this was not possible in some instances due to the dynamic nature of the clinical environment; therefore, a few subjects only had three samples drawn for co-oximetry.

To ensure that pulse oximetry values were stable prior to the co-oximetry blood draw, 2 minutes of sensor values were examined before each blood draw and the SD calculated for each of the sensors. If there was significant variation in saturation readings (determined by SD > 2) for the standard and the study sensors, the co-oximetry value was eliminated from analysis. We considered this degree of variation in pulse oximeter reading not a fair test of pulse oximetry performance, given its reliance on processing time. If a sensor did not have a recorded value at the time of the blood draw, no value was recorded for that sensor and that data was unavailable for analysis.

Statistical Analysis

An independent statistician who was paid by the authors for his analysis performed the analysis. He did not receive compensation from Masimo and did not have any contact with them as part of this study or analysis.

Demographic data were analyzed by calculating mean and SD for numerical variables and percentages for categorical variables.

Initial analyses focused on graphical comparisons of sensor measurements (Sp_{O_2}) with corresponding co-oximetry values (Sa_{O_2}) to assess patterns of disagreement. These included scatter plots, and smoothed regression estimates of average Sp_{O_2} levels plotted against corresponding Sa_{O_2} values. We also tabulated the proportion of Sp_{O_2} values that differed from corresponding Sa_{O_2} values by at least 5%, categorizing Sa_{O_2} values into three groups (≤ 75%, > 75–85%, and > 85%) corresponding to ranges of particular clinical interest.

Accuracy of the study sensors (Masimo Blue and Nellcor) Sp_{O_2} measurements was assessed using estimated bias, defined as the pairwise difference between the sensor Sp_{O_2} value and the corresponding Sa_{O_2} value measured at the same time point for

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each patient. Bias was summarized graphically and using Bland-Altman 95% limits of agreement, defined by the observed mean difference ± twice the observed SD of the observed differences (12). Because of possible within-individual correlation between successive measurements taken on a given patient, SDs were estimated using mixed-effect linear regression models for replicate measurements, including patient- and method-level effects (13). The resulting limits of agreement have a prediction interval interpretation and are generally somewhat wider than the standard Bland-Altman limits. Possible trends in mean differences and probability of greater than or equal to 5% differences with level of Sa_{O_2} were also evaluated with regression models, using generalized estimating equation methods to account for within-individual correlations in repeated measurements (14). Sa_{O_2} levels were treated both as continuous and categorical values, using the three categories based on ranges described above. Between-method differences were evaluated via test for interaction in these models. To investigate the possible influence of demographic factors on results, we extended regression models for mean and probability of greater than or equal to 5% differences to include age, height, weight, binary indicators of non-white race and female gender, and Massey Score (treated as a four-level categorical factor). To evaluate whether sensor placement in a standard location (thumb, finger, or great toe) versus a nonstandard location affected the results of the Masimo Standard, the average bias was compared between the two groups and *p* value obtained. Bland-Altman results were also obtained for these two groups.

TABLE 1. Demographics and Baseline Variables of Patients in This Study

Demographic or Baseline Variable	Patient Cohort, *n* = 50
Age (mo)	18 (0.8–83)
Weight (kg)	8.9 (3.5–17.1)
Male sex, *n* (%)	33 (66)
Race, *n* (%)	
White	37 (74)
Asian	10 (20)
African American	1 (4)
Massey Skin Color Scale	2.5 ± 0.8
Baseline oxygen saturation	78.5 ± 6
Diagnosis, *n* (%)	
Tetralogy of Fallot	25 (50)
Hypoplastic left heart syndrome	11 (22)
Tricuspid atresia	3 (6)
Double inlet left ventricle	3 (6)
Pulmonary atresia with intact ventricular septum	3 (6)
Other	5 (10)

RESULTS

Fifty patients were enrolled in the study, and 185 arterial blood samples were collected. The average age of the patients was 18 months (range, 25 d to 6 yr), average weight was 8.9 kg (range, 3.5–17.1 kg), baseline saturation as measured by the Masimo Standard sensor was 79% ± 6%, and the most common diagnosis was tetralogy of Fallot (**Table 1**). The breakdown showing locations of each sensor is shown in **Table 2**. Sensor placement for the Masimo Standard was determined by the anesthesiologist for clinical care, so there were many sensors placed not on the thumb or great toe, including placement on the hand, foot, and ear.

The average co-oximeter saturation was 81% with an SD of 8%, with 71% (n = 131) of the 185 samples having saturations less than or equal to 85%. Comparison of mean differences and corresponding variability (summarized by the SD and Bland-Altman limits of agreement) across the three sensors (**Table 3**) for all measurements indicated comparable mean values for the Masimo Blue and Nellcor sensors, with greater variability in measurements (SD) for the latter. Results from the Masimo Standard sensor indicated a slightly higher average mean difference and intermediate level of variability. Corresponding results for subsets of measurements defined by ranges of underlying Sa_{O_2} values indicated increasing levels of bias and variability for decreasing Sa_{O_2} levels for all three sensors, with the largest average differences occurring for Sa_{O_2} values less than 75%. Graphical displays of the Bland-Altman results for the Masimo Blue and Nellcor sensors (**Fig. 1**, ***A*** and ***B***) also illustrate the greater variability in measurements from the Nellcor, along with an apparent increase in variability at Sa_{O_2} levels less than 85%. (Note that although we chose to plot these results against Sa_{O_2} level rather than the mean of the paired measurements, our reported conclusions are similar for either choice.) The corresponding systematic increase in mean difference observed for decreasing Sa_{O_2} level for the Nellcor sensor was statistically significant via a test for linear trend ($p = 0.004$). This trend is particularly apparent for Sa_{O_2} levels less than 75% and was evaluated excluding one outlying Sa_{O_2} measure of 31%.

Based on our patient population of hypoxemic patients with congenital heart disease in whom we try to maintain saturations between 75% and 85%, we were particularly interested in the accuracy of the three sensors in patients with the saturation from pulse oximetry (Sp_{O_2}) in the ranges of less than or equal to 75%, 75–85%, and greater than 85%. **Table 4** presents estimated percentage of paired measurements for which the sensor Sp_{O_2} value differed from the corresponding Sa_{O_2} value by at least 5% for saturation values in these ranges. For the Masimo Blue sensor, 17% of all samples differed from co-oximetry values (higher or lower) by at least greater than or equal to 5%. Corresponding percentages were 25% and 29% for the Nellcor and Masimo Standard sensors, respectively. At saturation range of less than or equal to 85% and greater than 75%, 14% of the samples obtained from Masimo Blue, 24% of the readings from the Nellcor, and 31% from the Masimo Standard sensors were greater than or equal to 5% points difference. Percentage differences were higher for Sa_{O_2} values less than 75% for all three sensors. In addition, percentages for the Nellcor and Masimo Standard displayed increasing

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TABLE 2. Location of Sensor on the Body Broken Down by Sensor Type

Sensor Location	Masimo Blue, *n* (%)	Nellcor, *n* (%)	Masimo Standard, *n* (%)
Left great toe	11 (22)	17 (34)	7 (12)
Right great toe	10 (20)	6 (12)	16 (27)
Left thumb	12 (24)	17 (34)	7 (12)
Right thumb	16 (32)	8 (16)	5 (8)
Other	1 (2)	2 (4)	25 (42)
Finger	1 (2)	2 (4)	9 (15)
Non-great toe			1 (2)
Hand			7 (12)
Foot			6 (10)
Ear			2 (3)

trends with decreasing Sa_{O_2} level, representing more inaccuracy at lower saturations. This trend was statistically significant for the Nellcor sensor ($p = 0.004$), but not for the Masimo Standard ($p = 0.063$). An overall test for differences between methods in the patterns of percentages by category of Sa_{O_2} was significant ($p = 0.03$).

Additional analyses investigating the possible impacts of demographic factors on conclusions did not reveal any significant findings for either the Nellcor or the Masimo Standard sensor. Models for the mean bias from the Masimo Blue sensor yielded a significant effect for a Massey Score of 4 ($p = 0.006$), indicating an increase in average bias of 4% relative to individuals with a score of 1, adjusting for other demographic factors in the model. However, this effect was not significant ($p = 0.33$) in the corresponding model for the probability of bias of at least 5%.

Subset analysis was performed on the Masimo Standard to evaluate whether nonstandard location placement affected the results. **Table 5** shows the mean bias between Sp_{O_2} and Sa_{O_2} for the Masimo Standard sensor, broken down by location of sensor placement. There was a difference in the mean bias, but it was not significant with a *p* value of 0.33. The Bland-Altman results were also performed for these subsets, limits for the standard placement were −7.4 and 11.8, and the nonstandard was −7.1 and 14.2. In comparison, the Bland-Altman limits of the entire data set were −7.3 and 12.2.

DISCUSSION

Pulse oximetry is fundamental to the management of children with pathologies of oxygen delivery, and clinicians may rely on it without fully understanding its limitations, particularly at lower saturations. In this investigator-initiated study, we examined the performance of three different pulse oximeters in 50 infants and children with congenital cardiac disease and baseline saturations below 90%. All three sensors demonstrated more bias at saturations less than 85% (the target saturations of this patient population), consistent with prior studies looking at the accuracy of pulse oximetry (6). All three sensors were also imprecise, based on the two SDs of the difference of $Sa_{O_2} - Sp_{O_2}$ being greater than 5%. As a result of this imprecision, clinical decisions should not be based solely on pulse oximetry as one single Sp_{O_2} value may over- or underestimate the true Sa_{O_2} value.

A main goal of this study was to determine which of the three sensors had the best performance in the saturation range of 75–85%, a clinically vital question because many patients with congenital heart defects have values that are regularly in this range. The Masimo Blue sensor and the Nellcor sensor had less bias at saturations 75–85% than the Masimo Standard sensor. The Masimo Blue sensor had improved precision at saturations 75–85% compared with the other two sensors, and the accuracy of the Masimo Blue sensor was within 5% points of the gold-standard Sa_{O_2} the majority of the time.

We also separately analyzed saturations less than 75% and found in this range an increased number of samples with greater than or equal to 5% difference from the gold standard, which could confound clinical decision making. In cyanotic children, clinical decisions are frequently made based on the saturation displayed by the pulse oximeter. From a decision in clinic about whether to admit a patient for closer surveillance or a decision to change a surgical date, to acute decisions in the ICU or OR about when to get an arterial blood gas to confirm the saturation value, pulse oximetry can have a profound impact on patient care. For saturations under 80%, a small drop in Pa_{O_2} results in a larger saturation decrease than at higher saturations.

For this study, we defined less than or equal to 5% points difference in pulse oximetry reading (Sp_{O_2}) and the Sa_{O_2} (co-oximetry gold standard) between 75% and 85% as being a clinically acceptable range for managing patient care. Previously published guidelines for neonatal screening of heart disease use a 5% points change in pulse oximetry to alter

TABLE 3. Mean Paired Differences Between Estimated Oxygen Saturation From Pulse Oximetry and Arterial Blood Oxygen Saturation, and Bland-Altman Limits of Agreement for All Three Sensors

	All Sa_{O_2} Values		$Sa_{O_2} \leq 75\%$		$> 75\%$ $Sa_{O_2} \leq 85\%$		$Sa_{O_2} > 85\%$	
Sensor Type	Mean (SD)	BA Limits	Mean (SD)	BA Limits	Mean (SD)	BA Limits	Mean (SD)	BA Limits
Masimo Blue	1.7 (3.3)	−5.0, 8.4	2.1 (4.7)	−7.3, 11.5	1.9 (2.9)	−3.9, 7.6	0.9 (3.1)	−5.2, 7.0
Nellcor	1.7 (5.4)	−9.2, 12.5	4.4 (8.4)	−12.3, 21.2	1.5 (4.0)	−6.4, 9.4	−0.8 (3.8)	−8.5, 6.8
Masimo Standard	2.4 (4.9)	−7.3, 12.2	3.6 (6.2)	−8.7, 16.0	2.8 (4.6)	−6.5, 12.0	1.0 (4.2)	−7.4, 9.4

Sa_{O_2} = arterial blood oxygen saturation, BA = Bland-Altman.

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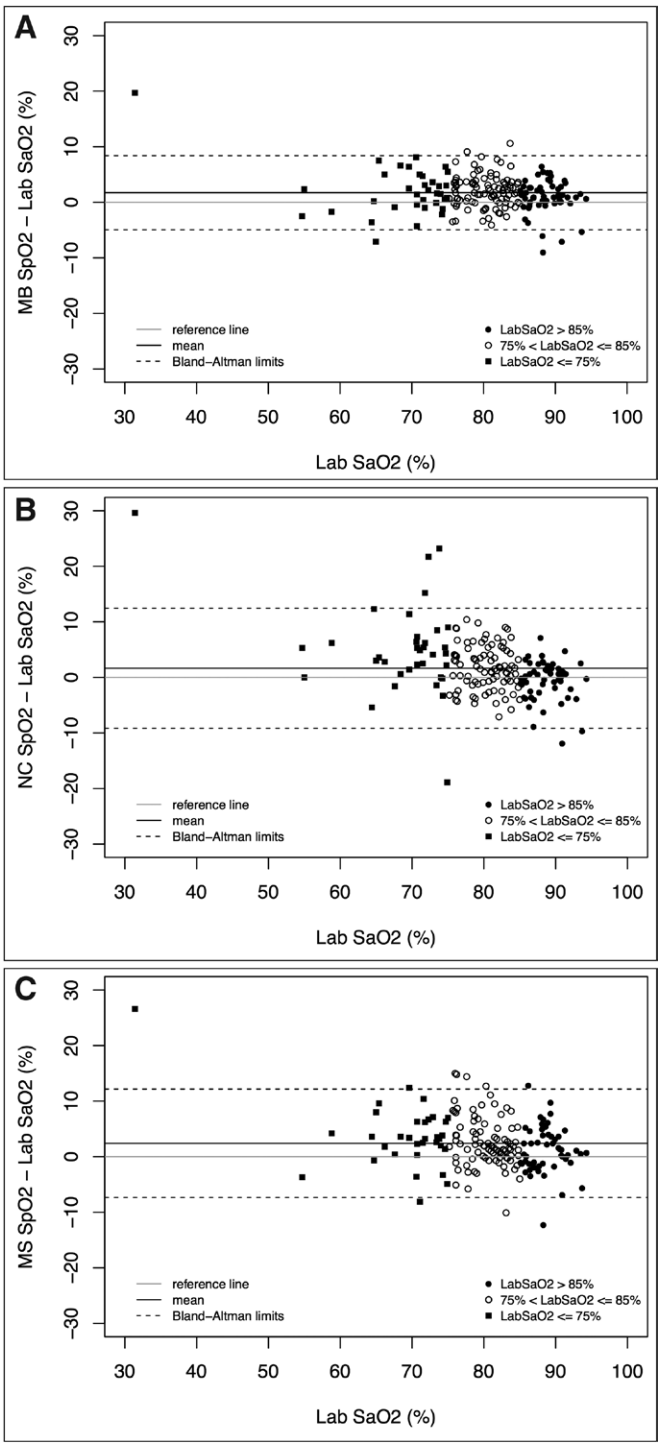

Figure 1. A, Bland-Altman limits and mean difference between Masimo Blue (MB) Spo_2 and Sao_2 plotted as a function of Sao_2 level. **B**, Bland-Altman results for the Nellcor (NC) sensor. **C**, Bland-Altman results for the Masimo Standard (MS) sensor.

clinical management (10). Similar use of 5–7% pulse oximetry changes as motivation for administering oxygen has been well established in pediatric respiratory diseases such as respiratory syncytial virus infection and bronchiolitis (11); however, it is important to note that this difference is at saturations over 85%. Based on these studies and our clinical practice, we classified Spo_2 values that differed by 5% from Sao_2 as acceptable for clinical management. At saturations below 75%, the true clinically acceptable range of difference may be even less than 5%; at these saturation levels, even small changes in saturations could result in larger changes in oxygen delivery.

We found that all the sensors studied had readings with a greater than or equal to 5% points difference across all of the study saturation ranges: the Masimo Blue sensor had 17% of its readings with a greater than or equal to 5% points difference; the Nellcor 25%; and Masimo Standard 29%. The device manufacturers report only a 3–4% difference from co-oximetry at saturations above 80%. Hence, it is unclear why a significant proportion of the samples in this study have more variation than the ranges provided by the device manufacturers; the sensors were used as specified and in a typical clinical environment. It is possible that the dynamic environment of actual clinical care, compared with stable environment in volunteers used for device testing, contributed to these increased differences. While these devices are tested for U.S. Food and Drug Administration approval, even when specified for use in neonatal patients, manufacturers can primarily test them on adults with only small studies performed in the target population (15).

The Masimo Blue sensor is approximately twice the cost of the Masimo Standard sensor. Although the Masimo Blue sensor is clearly more accurate than the Masimo Standard at the

TABLE 4. Sensor Measurements Greater Than or Equal to 5% Different Than Corresponding Arterial Blood Oxygen Saturation Values

Method	All Sao_2 Values % (*n*)	Sao_2 ≤ 75% % (*n*)	75% < Sao_2 ≤ 85% % (*n*)	Sao_2 > 85% % (*n*)
Masimo Blue	17 (185)	29 (35)	14 (95)	16 (55)
Nellcor	25 (182)	51 (35)	24 (91)	11 (56)
Masimo Standard	29 (183)	36 (33)	31 (94)	21 (55)

Sao_2 = arterial blood oxygen saturation, *n* = no. of paired measurements.

TABLE 5. Mean Paired Differences Between Estimated Oxygen Saturation From Pulse Oximetry and Arterial Blood Oxygen Saturation and Bland Altman Limits of Agreement for the Masimo Standard Sensor, Broken Down Into Groups Based on Sensor Placement

Sensor Placement	Mean Bias	Bland-Altman Limits
Standard (*n* = 144)	2.3	−7.4, 11.8
Nonstandard (*n* = 42)	3.4	−7.1, 14.2
All (*n* = 186)	2.4	−7.3, 12.2

Standard locations: thumb, finger, or great toe.

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lower saturations, it is important to evaluate which patients might most benefit from use of these sensors given their additional cost. Our data support the use of Masimo Blue sensor for patients with saturations below 85% because of the improved accuracy; however, it is still not completely reliable. Furthermore, those over 85% would not benefit, as the standard less-expensive sensors would be adequate in this range.

Although significantly larger than prior studies on the Masimo Blue and Nellcor sensors, this study includes a sample size of only 50 patients, which is a limitation of the study. This sample size did not allow for additional subgroup analysis. The OR and catheterization laboratory, while providing convenient arterial access, are areas where patient status fluctuated frequently, making it difficult to collect steady-state samples. Another limitation was that the sensor location was not controlled for the Masimo Standard, as part of clinical care it was placed by the anesthesiologist, and as a result study sensor location could not be randomized. The Masimo Standard sensor was at times placed in nonstandard locations, which could have affected the results. However, a subset analysis of the bias based on standard versus nonstandard location did not show a significant difference. The mean bias and the Bland-Altman limits of the sensors placed in the standard location were very similar to the results obtained with nonstandard location placement. In addition, this was an observational study, and a robust cost analysis was not possible because only the standard sensors were used for clinical decision making.

Our results suggest that use of a specially designed pulse oximeter for monitoring lower saturation levels, such as the Masimo Blue, has a limited role in children with cyanotic heart disease who have saturations consistently less than 85%. Although specifically designed for the cyanotic patient, the Masimo Blue sensor has imprecision at this range and therefore clinical decision cannot be based solely on the results of this noninvasive technology. This study highlights the ongoing need for additional development by the device manufacturers with calibration and development of algorithms for pulse oximeters with improved overall accuracy (bias and precision) at saturations less than 85% before it can be relied on for clinical decision making. At saturations less than or equal to 75%, with all sensors there are even more significant inaccuracies. Based on this study, it is our opinion that clinical decisions at saturations below 85% should not be made by pulse oximetry alone; instead, co-oximetry and/or an arterial blood gas is warranted.

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